MATH 345 Skeleton Notes

Unit 6: Constraints and SVD

Section 6.1: Absolute extrema on closed bounded regions

Recall what it means for a set to be *closed* from Closed sets.

Activity

Determine whether each of the given regions is closed.

Task The region $A = \{(x, y) : x^2 + y^2 < 1\}$.

Task The region $B = \{(x, y) : x^2 + y^2 = 1\}$

Task $C = \{(x, y) : 1 \leq x \leq 2, 1 \leq y \leq 2\}$

Task The set $D = \{(x, y) : x > 1\}$

Theorem If f is a scalar-valued function which is continuous on a closed and bounded set D, then f has an absolute maximum and an absolute minimum value on D.

The second derivative test informs us whether a critical point is a local maximum, a local minimum, or a saddle point. However, it doesn't identify global (absolute) extrema. For functions defined on closed, bounded regions, we can use the following procedure to find absolute extrema, provided that the boundary points of the region are of a suitably simple nature.

Given a function f, continuous on a closed bounded set D, this algorithm finds the absolute extrema of f on D.

1. Find all critical points of f in the interior of D.
2. Find all boundary candidates. On each boundary piece, reduce to a one-variable problem when possible, and include endpoints and corners.
3. Evaluate f at every candidate from the first two steps.
4. The largest value is the absolute maximum value, and the smallest value is the absolute minimum value.

If the boundary of D is made from one or more curves, we can usually parameterize each curve and find boundary candidates using one-variable calculus. Endpoints and corners must still be checked separately.

Activity

Consider the function

$$f(x, y) = 3x^2 - 2y^2 + 2x + 2y - 1$$

on the unit square

$$D = \{(x, y) \in \mathbb{R}^2 \,:\, -1 \le x \le 1, -1 \le y \le 1\}.$$

Find the absolute maximum and minimum values of f on D.

Activity

In this exercise, we compute the absolute extrema of the function $f(x, y) = 2x - 2xy + y^2$ for $0 \le x \le 4$ and $0 \le y \le 3$.

Task Start by finding all critical points on the interior of the domain upon which we are trying to optimize f.

Task Next, find the maximum and minimum values on the interior of the boundary curves of the region, and on the corners of the region.

Task Finally, compare all values.